Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/09/2018
PI Name:	Riaz, Amir Ph.D.		
Project Title:	Gravity Scaling of Pool Boiling Heat Transfer: Numerical Simulations and Validation with MABE and NPBX		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
Start Date:	08/16/2016	**End Date:**	08/18/2020
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/18/2020 per NSSC information (Ed., 11/12/19) NOTE: Extended to 8/15/2019 per NSSC information (Ed., 10/3/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kim, Jungho Ph.D. (University of Maryland) Balaras, Elias Ph.D. (The George Washington University)		
Grant/Contract No.:	NNX16AQ77G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The key challenge in developing two-phase thermal systems for space environments is the development of a heat transfer database and reliable models for boiling in variable gravity environments from which the performance of two-phase heat exchangers can be predicted. The Microheater Array Boiling Experiment (MABE) and Nucleate Pool Boiling eXperiment (NPBX) were flown on the International Space Station (ISS) in 2011 within the Boiling Experiment Facility (BXF). MABE provided local heat transfer and visualization of pool boiling heat transfer behavior in a microgravity environment. Earth gravity data obtained prior to launch served as the reference data. The ability of a model developed from parabolic aircraft work and ISS data to predict how boiling was altered under a range of gravity levels was tested and found to predict the ISS data within 20%. The focus of NPBX was the investigation of how single and multiple bubbles (up to 5) grew and merged in microgravity. Heat fluxes for steady state nucleate boiling and critical heat flux were found to be much lower than those obtained under Earth gravity. It was found that heater size, chamber geometry, and bulk fluid conditions must be accounted for in nucleate boiling heat transfer correlations under microgravity conditions. In the proposed work, high fidelity numerical simulations of the boiling process will be used to glean further insights into the boiling process. The data from MABE and NPBX will be used to validate the simulations as they provide a wide range conditions (heater sizes, subcoolings, and superheats including CHF) at which to test the numerical results. NPBX will serve as validation for simulations with a few nucleation sites in microgravity. Simulations of boiling in Earth gravity will be made for the MABE heater and compared with the MABE Earth gravity results, after which additional simulations will be performed as gravity is slowly decreased. The results will be compared with the MABE microgravity results and the model. Co-Investigators for the proposed work are Prof. Elias Balaras (George Washington University) and Prof. Jungho Kim (U. of Maryland, PI for MABE).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reliable, physics based direct numerical simulation techniques that are also numerically efficient will be used to interrogate the fundamental physics of the boiling process under low-gravity and Earth-gravity conditions. Such high accuracy modeling of the boiling process will help develop effective heat transfer methodologies for the cooling of various engineering equipment including microprocessors.
Task Progress:	Our research to date has made following contributions which are important for accurate numerical simulation of pool boiling. The contributions include: 1. Development of a parallel, MPI-based fully three-dimensional solver that uses adaptive mesh refinement (AMR) and state-of-the-art methods for resolving the physics of phase-change process involved in pool boiling. 2. Development of a coupled microlayer - dynamic contact line model to account for inertial effects on the heater surface during bubble mergers. 3. Grid independence and accuracy study of the aforementioned model in Earth gravity and microgravity. 4. Development of a nucleation site model to mimic experimental bubble density. 5. High fidelity simulations of subcooled pool boiling to understand effect of gravity on heat transfer in the BDB (buoyancy dominated boiling), SDB (surface tension dominated boiling), and the transition regime.
Bibliography Type:	Description: (Last Updated: 04/23/2026)
Abstracts for Journals and Proceedings	Dhruv A, Balaras E, Riaz A, Kim J. "Level set immersed boundary method for gas-liquid-solid interactions with phase-change." 70th Annual Meeting of the APS Division of Fluid Dynamics, Denver, Colorado November 19-21, 2017. Bulletin of the American Physical Society. 2017 Nov;62(14):BAPS.2017.DFD.A7.1. http://meetings.aps.org/link/BAPS.2017.DFD.A7.1 , Nov-2017